

AI Data Scientist Report

1. Dataset Overview

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2. Model Performance

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3. Model Explainability (SHAP)

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4. AI Business Analyst Insights

{'type': 'text', 'text': "EXECUTIVE SUMMARY

We analyzed customer data from 7,043 accounts to identify the primary drivers of customer retention and churn. Our predictive models achieved high accuracy, with Logistic Regression leading the performance at 83.75 percent. The analysis reveals that customer demographics, specifically family status, and contract types are the strongest predictors of customer behavior. These insights provide a clear path to optimize our retention strategies and maximize customer lifetime value.

KEY DATA INSIGHTS

- The average customer tenure is approximately 32 months, with a wide distribution ranging from brand new customers to long-term accounts of 72 months.
- Monthly charges average 64.76 dollars, with a standard deviation of 30.09 dollars, indicating a highly diverse product mix and pricing structure.
- Half of our customer base has a tenure of under 29 months, highlighting a critical window for early-stage retention efforts.

MODEL PERFORMANCE

We evaluated three machine learning models to predict customer behavior. Logistic Regression

emerged as the top-performing model with an accuracy of 83.75 percent. Random Forest followed closely at 83.25 percent, while XGBoost achieved 80.27 percent. The high accuracy of the Logistic Regression model ensures reliable, actionable predictions for our marketing and customer success teams.

IMPORTANT DRIVERS

Customer family status is the most influential factor, with the presence or absence of dependents having the highest impact on the model's predictions. Customer tenure and relationship status, specifically having no partner, also play critical roles in determining loyalty. Additionally, contract structures heavily influence outcomes, where long-term two-year contracts promote stability, while month-to-month contracts increase risk. Financial factors like monthly charges and payment methods further refine our understanding of customer risk profiles.

BUSINESS RECOMMENDATIONS

- Design targeted family-centric promotions and multi-line discounts to attract and retain customers with dependents.
- Implement proactive engagement campaigns during the first 9 to 12 months of tenure to transition high-risk, short-term customers into longer-term relationships.
- Incentivize month-to-month customers to migrate to two-year contracts by offering loyalty rewards or small monthly discounts.

FINAL CONCLUSION

By leveraging our top-performing predictive model, we can proactively identify at-risk customers and deploy targeted retention strategies. Focusing on family demographics, early tenure engagement, and contract optimization will significantly reduce churn and stabilize long-term revenue."

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